# BIO1

General Biology 1

Grade 12, Semester 1 study outline

*A summary of section headings and key terms, made by students for revision. It is not the handout and does not replace it.*

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## 01. Cell Theory

### Cell Theory

### I. Introduction

### II. Timeline of Cell Theory

- 1665- Robert Hooke - He coined the term cell. - He used dried cork from a compound microscope.
- 1675- Anton Van Leeuwenhoek - He was the first to observe free-living cells. - He was also the first to describe the sperm cells of humans, dogs, rabbits, frogs, and insects - He was the first to observe the blood cells of mammals, birds, amphibians, and fish.
- 1838 - Theodor Schwann and Mattias Schleiden - Schleiden defined the cell as the basic unit of plant structure, and a year later, Schwann defined the cell as the basic unit of animal structure. - Each conjectured that there was a nucleus.
- 1855- Rudolf Virchow - Stated that 'Where a cell exists, there must have been a pre- existing cell' - Developed the third tenet of the cell theory

### III. The Cell Theory

- -- 1 of 2 -- There are three (3) postulates of cell theory.

### 1. All living things are made up of cells

- Cells are the fundamental building blocks used to create tissues, organs, and entire functioning organisms.

### IV. Exceptions to the Cell Theory

### 1. Set of Genes

### 2. Plasma membrane

### 3. Metabolic Machinery

### 1. They have genetically determined macromolecular organization

### 2. They have genetic hereditary material in the form of DNA or RNA

### 3. A capacity of auto-reproduction

### 4. A capacity for mutation

### V. Unlocking the terms

- Cell theory- It is the historical scientific theory that unlocks all scientific breakthroughs in cells.

### B. DNA (deoxyribonucleic acid)

- - It is a double-stranded organic chemical that contains genetic information that is passed onto offspring.

### C. RNA ( ribonucleic acid )

- -It is a molecule similar to DNA.
- It is a molecule essential in various biological roles in the coding, decoding, regulation, and expression of genes.
- Macromolecular organization- it is the arrangement of molecules that are covalently bonded such as proteins, carbohydrates, and nucleic acids.
- Mutation- It is a change that occurs in our DNA sequence, either due to mistakes when the DNA is copied.

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## 02. Cell Structure and Functions

### Cell Structure and Functions

### I. Introduction

- According to the cell theory plants and animals are made up of the unit called cells.

- As a living organism, we are composed of cells, but what are the certain structures that make cells?

### II. What are cells?

- The actual word, cell, in turn, comes from Latin and means "a chamber." This terminology arises from the early observation of dead cork under the microscope.
- When viewed through the microscope, dead cork is shown to consist of a complex lattice work of hollow chambers with stiff walls.
- Each dark, hollow chamber in cork bark, then, is a type of non-living cell.
- But in general, however, the cell is defined as a living, tiny chamber that is surrounded by a thin membrane and contains various organelles.

### Microscopic view of cells

### III. Eukaryotic cells

### IV. Parts of a cell and its function

### B. RIBOSOMES- protein factories

- -They are composed of ribonucleic acid (RNA) and proteins and they are the sites of protein synthesis. -- 2 of 7 --

### C. ENDOPLASMIC RETICULUM- highway

- -is literally a "tiny network present within the cytoplasm." - It is a complex network of flattened sacs that serves to carry things around in the cell, much like a miniature circulation or highway system.

### Rough ER- contains many ribosomes

### Smooth ER- do not have ribosomes

### D. GOLGI APPARATUS- receiving,sorting, shipping center of the cell

- -We can think of the Golgi as a warehouse for receiving, sorting, shipping, and even some manufacturing. - It is where the products of the ER, such as proteins, are modified and stored and then sent to other destinations. -- 3 of 7 --

### E. LYSOSOMES- digestive compartment

- -contains digestive enzymes which, when released, digest or break down foodstuffs and other materials within the cell

### F. VACUOLE- compartment

- - it is a storage sac for various digestive enzymes. -- 4 of 7 -- Note: Plant cells have a larger vacuole than animal cells.

### G. CYTOPLASM- refers only to the region between the nucleus and the plasma membrane

### I. CELL WALL (present only in Plant Cells)

- -- 5 of 7 -- - outer layer that maintains cell's shape and protects cell from mechanical damage; made of cellulose, other polysaccharides, and protein.

### J. PEROXISOME- detoxify alcohol and other harmful compounds by transferring hydrogen from the

### K. MITOCHONDRIA- powerhouse of the cell

- -they contain the chemicals necessary for aerobic respiration and ATP production. -- 6 of 7 -- L.

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## 03. Prokaryotic and Eukaryotic Cells

### Prokaryotic and Eukaryotic Cells

### I. INTRODUCTION

- The organisms with only one cell in their body are called unicellular organisms (e.g., bacteria, blue green algae, some algae, Protozoa, etc.).
- The organisms having many cells in their body are called multicellular organisms (e.g., most plants and animals).

### II. Types of cells

### A. PROKARYOTIC CELLS

- The prokaryotic cells are the most primitive cells from the morphological point of view.
- Examples of prokaryotic cells: mycoplasma, bacteria and cyanobacteria or blue algae.

### B. EUKARYOTIC CELLS

- Examples of Eukaryotic cells: protists, plants, animals and fungi They HAVE DNA, Ribosomes, Cytoplasm, Cell Membrane and SOME HAVE cell walls. (Note: Cell walls are present in plants and fungi only, not in animals).

### REMEMBER

- Example: A bacterium is a PROKARYOTE.
- Example: A dog is a EUKARYOTE and it has EUKARYOTIC

### CELL

### III. ENDOSYMBIOTIC THEORY

- -- 1 of 3 -- The endosymbiotic theory states that some of the organelles in today's eukaryotic cells were once prokaryotic microbes.

### IV. Unlocking the terms

- Fig.1- homologous structures Histones- are a family of basic proteins that associate with DNA in the nucleus and help condense it into chromatin. -- 2 of 3 -- Fig 2.- Chromosomes and Histones Chromosomes- are thread-like structures located inside the nucleus of animal and plant cells.

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**Key terms**

Histones - family of basic proteins that associate with DNA in the nucleus and help condense it into

Chromosomes - thread-like structures located inside the nucleus of animal and plant cells

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## 04. Cell Types

### Cell Types

### I. Introduction

- Cells form a working animal body through their emergent properties, which arise from successive levels of structural and functional organization.
- The atom is the smallest and most fundamental unit of matter.
- It consists of a nucleus surrounded by electrons.
- A molecule is a chemical structure consisting of at least two atoms held together by one or more chemical bonds.
- Many molecules that are biologically important are macromolecules, large molecules that are typically formed by polymerization (a polymer is a large molecule that is made by combining smaller units called monomers, which are simpler than macromolecules).
- Some cells contain aggregates of macromolecules surrounded by membranes; these are called organelles.
- All living things are made of cells; the cell itself is the smallest fundamental unit of structure and function in living organisms. (This requirement is why viruses are not considered living: they are not made of cells. -- 1 of 5 -- In larger organisms, cells combine to make tissues, which are groups of similar cel...
- An organ system is a higher level of organization that consists of functionally related organs.

### II. TYPES OF CELL TISSUES

- EPITHELIAL TISSUE- This type of tissue is commonly seen outside the body as coverings or as linings of organs and cavities.
- Epithelial tissues are characterized by closely-joined cells with tight junctions (i.e., a type of cell modification).
- The matrix generally consists of a web of fibers embedded in a liquid, jellylike, or solid foundation.

### TYPES OF CONNECTIVE TISSUES

- Adipose tissues are also examples of loose connective tissues that store fats which function to insulate the body and store energy.
- Chondrocytes are the cells that secrete collagen and chondroitin sulfate.
- MUSCLE TISSUE - These tissues are composed of long cells called muscle fibers that allow the body to move voluntarily or involuntary.
- Movement of muscles is a response to signals coming from nerve cells. -- 4 of 5 --

### CATEGORIES OF MUSCLE TISSUES

- - skeletal-striated; voluntary movements - cardiac-striated with intercalated disk for synchronized heart contraction; involuntary - smooth-not striated; involuntary D.
- NERVOUS TISSUE - These tissues are composed of nerve cells called neurons and glial cells that function as support cells.
- The dendrite is the part of the neuron that receives impulses from other neurons while the axon is the part where the impulse is transmitted to other neurons.

**Key terms**

Organelles - small structures that exist within cells

Organs - collections of tissues grouped together performing a common function. Organs are present not only in

## 05. Cell Modifications

### Cell Modifications

### I. Introduction

- Organ systems include specialized organs made up of specialized tissues and cells.

### II. What is Cell Modification?

- Cell modification Is the change in the structure of a cell to carry out specific functions.

### III. Types of Cell Modification

### A. Apical Cell Modification- (Surface or luminal)

- The word "flagellum" means "whip".
- Some special flagella are used in few organisms as sensory organs that can sense changes in pH and temperature.

### 3. Microvilli- are microscopic cellular membrane protrusions that increase the

### 4. Pseudopods - is a temporary arm-like projection of a eukaryotic cell membrane that is

- Filled with cytoplasm, pseudopodia primarily consist of actin filaments and may also contain microtubules and intermediate filaments.
- Because it is the surface closest to the underlying blood supply, it often contains receptors for blood borne factors such as hormones.
- The best examples are found in the basal layers of stratified squamous epithelium.
- Several types of junctions can be seen, such as: 1.
- It is a closable channel that -- 2 of 4 -- connects the cytoplasm of adjoining animal cells.

### IV. Coordination and Control

**Key terms**

Hemidesmosomes - located on the inner surface of basal plasma membranes in contact with the basal

## 06. Cell Cycle (Mitosis)

### Cell Cycle (Mitosis)

### I. Introduction

- The ability of organisms to produce more of their own kind is the one characteristic that best distinguishes living things from nonliving matter.
- In 1855, Rudolf Virchow, a German physician, put it this way: "Where a cell exists, there must have been a preexisting cell, just as the animal arises only from an animal and the plant only from a plant." He summarized this concept with the Latin saying "Omnis cellula e cellula," meaning "Every cell from a cell." Th...

### II. Cell Division

- Cell division is an integral part of the cell cycle, the life of a cell from formation to its own division.

### A. Most cell division results in genetically identical daughter cells

- In both prokaryotes and eukaryotes, most cell division involves the distribution of identical genetic material-DNA-to two daughter cells

### B. Cellular Organization of the Genetic Material

- A cell's DNA, its genetic information, is called its genome.
- The replication and distribution of so much DNA are manageable because the DNA molecules are packaged into structures called chromosomes (from the Greek chroma, color, and soma, body) The entire complex of DNA and proteins that is the building material of chromosomes is referred to as chromatin.
- Every eukaryotic species has a characteristic number of chromosomes in each cell nucleus -- 1 of 6 -- Somatic cells (non reproductive cells) have two sets of chromosomes Gametes (reproductive cells: sperm and eggs) have half as many chromosomes as somatic cells Eukaryotic chromosomes consist of chromatin, a complex ...

### C. Distribution of Chromosomes during Eukaryotic Cell Division

- In preparation for cell division, DNA is replicated and the chromosomes condense Each duplicated chromosome has two sister chromatids, which separate during cell division The centromere is the narrow "waist" of the duplicated chromosome, where the two chromatids are most closely attached

### III. Eukaryotic Cell Division

- Eukaryotic cell division consists of: - Mitosis, the division of the nucleus - Cytokinesis, the division of the cytoplasm Gametes are produced by a variation of cell division called meiosis Meiosis yields nonidentical daughter cells that have only one set of chromosomes, half as many as the parent cell

### IV. Phases of Cell Cycle

- The cell cycle consists of Mitotic (M) phase (mitosis and cytokinesis) and Interphase (cell growth and copying of chromosomes in preparation for cell division).

### 1. Prophase

### Chromosomes condense and become visible

### Golgi and ER are dispersed

### Nuclear envelope breaks down

### 2. Prometaphase

### Chromosomes attach to microtubules at the kinetochores

- -- 3 of 6 -- Each chromosome is oriented such that the kinetochores of sister chromatids are attached to microtubules from opposite poles.

### Chromosomes move to equator of the cell

### 3. Metaphase- M for middle

- - All chromosomes are aligned at equator of the cell, called the metaphase plate - Chromosomes are attached to opposite poles and are under tension

### 4. Anaphase- A for away

- -- 4 of 6 -- Proteins holding centromeres of sister chromatids are degraded, freeing individual chromosomes

### Chromosomes are pulled to opposite poles

### Spindle poles move apart

### 5. Telophase

### Chromosomes are clustered at opposite poles and decondense

### Nuclear envelopes reform around chromosomes

### Golgi complex and ER re-form

### Spindle is disassembled

### Cytokinesis is well underway by late telophase

- - In animal cells, cleavage furrow forms to divide the cells - In plant cells, cell plate forms to divide the cells

### V. Abnormal Cell Cycle: Cancer

- Cancer is the uncontrolled growth and division of cells.

### Causes of Cancer

- Substances that are known to cause cancer are called carcinogens (kar SIH nuh junz).
- Tobacco, tobacco smoke, alcohol, some viruses, and radiation are examples of carcinogens What to do: Prevention Avoiding carcinogens can help reduce the risk of cancer.

### VI. Apoptosis

- Apoptosis (a pup TOH sus) is a natural process of programmed cell death.

#### Key terms

Cell division - integral part of the cell cycle, the life of a cell from formation to its own division

Gametes - produced by a variation of cell division called meiosis

Cytoskeleton - disassembled: spindle begins to form

Chromosomes - clustered at opposite poles and decondense

Cytokinesis - well underway by late telophase

Cancer - uncontrolled growth and division of cells. Cancer cells grow and divide as long as they receive

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## 07. Cell Cycle (Meiosis)

### Cell Cycle (Meiosis)

### I. Cell Division by Meiosis: The Basis of Sex

- These cells are produced by a form of cell division called meiosis.

### Haploid Cells

- Gametes or sex cells are the most common type of haploid cells.

### Diploid Cells

### II. Meiosis Stages

### A. Meiosis I reduces the chromosome number

- Once paired, the sister chromatids of each chromosome are visible, showing that each consists of two sister chromatids joined by a single centromere.

### B. Meiosis II begins with haploid cells

- In prophase II, the unpaired chromosomes consist of two sister chromatids joined by a centromere.

### III. Comparing Mitosis and Meiosis

### V. Unlocking Vocabulary

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## 08. Transport Mechanisms (Simple Diffusion and Facilitated Diffusion)

### I. Introduction

- The ability of the cell to discriminate in its chemical exchanges is fundamental to life, and it is the plasma membrane and its component molecules that make this selectivity possible.

### II. Cellular Membranes are fluid mosaics of lipids and proteins

### A. Phospholipid bilayer

- - Every cell membrane is composed of phospholipids in a bilayer.

### B. Transmembrane proteins

- - A major component of every membrane is a collection of proteins that float in the lipid bilayer. - These proteins have a variety of functions, including transport and communication across the membrane.

### C. Interior protein network

### D. Cell-surface markers

- - membrane sections assemble in the endoplasmic reticulum, transfer to the Golgi apparatus, and then are transported to the plasma membrane. - Different cell types exhibit different varieties of these glycoproteins and glycolipids on their surfaces, which act as cell identity markers

### IV. Cellular membranes have an organized substructure

- - It was believed that because of its fluidity, the plasma membrane was uniform, with lipids and proteins free to diffuse rapidly in the plane of the membrane.

### V. Evolution of Plasma Membrane

### A. Freshwater and Saltwater fish

### B. Bacteria and Archaea

- -thrive at temperatures greater than 90°C (194°F) in thermal hot springs and geysers.
- Their membranes include unusual lipids that may prevent excessive fluidity at such high temperatures.

### VI. Summary

### Component Composition Function How it Works Example

### Phospholipid layer

### Phospholipid

### Molecules

### Provides permeability

### Excludes water-soluble

### Bilayer of cell is

### Transmembrane

### Proteins

### Carriers

### Actively or Passively

### Move specific molecules

### Glycophorin carrier for sugar

**Channels**
**Passive transport**
**Create a selective tunnel that**
**Sodium and potassium**
**Receptors**
**Transmit information**

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## 09. Transport Mechanisms (Simple Diffusion and Facilitated Diffusion)

**I. Proteins: Multi-functional Components**

**II. Proteins and protein complexes perform key functions**

- Most cell types carry their own ID tags, specific combinations of cell-surface proteins and protein complexes such as glycoproteins that are characteristic of that cell type.

**III. Structural features of membrane proteins relate to function**

**A. The anchoring of proteins in the bilayer**

**B. Transmembrane Domains**

- These domains are composed of hydrophobic amino acids usually arranged into helices.

**C. Pores**

- This so-called barrel, open on both ends, is a common feature of the porin class of proteins that are found within the outer membrane of some bacteria.

**IV. Passive Transport Across the Membrane**

- -- 3 of 4 -- Many substances can move in and out of the cell without the cell's having to expend energy.
- This type of movement is termed passive transport.

**A. Transport can occur by simple diffusion**

**B. Proteins allow membrane diffusion to be selective**

- These molecules can still enter the cell by diffusion through specific channel proteins or carrier proteins embedded in the plasma membrane, provided there is a higher concentration of the molecule outside the cell than inside.
- This process of diffusion mediated by a membrane protein is called facilitated diffusion.

**C. Facilitated diffusion of ions through channels**

- Therefore, ions cannot move between the cytoplasm of a cell and the extracellular fluid without the assistance of membrane transport proteins.

**D. Facilitated diffusion by carrier proteins**

- - Carrier proteins can help transport both ions and other solutes, such as some sugars and amino acids, across the membrane.
- Transport through a carrier is still a form of diffusion and therefore requires a concentration difference across the membrane. - Carriers must bind to the molecule they transport, so the relationship between concentration and rate of transport differs from that due to simple diffusion. - As concentration increases,...
- But when a carrier protein is involved, a concentration increase means that more of the carriers are bound to the transported molecule.
- This means that the carrier exhibits saturation.

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## 10. Transport Mechanisms (Active Transport and Bulk Vesicular Transport)

**I. Osmosis**

- This net diffusion of water across a membrane toward a higher solute concentration is called osmosis.
- When two solutions have the same osmotic concentration, the solutions are isotonic (Greek iso, "equal"). -- 1 of 4 --

**II. How solutes create osmotic pressure?**

**III. Maintaining osmotic balance**

- Organisms have developed many strategies for solving the dilemma posed by being hypertonic to their environment and therefore having a steady influx of water by osmosis:

**A. Extrusion**

### C. Turgor

### IV. Active Transport Across Membranes

- This process requires the expenditure of energy, typically from ATP, and is therefore called active transport.

### A. Active transport uses energy to move materials against a concentration gradient

- Like facilitated diffusion, active transport involves highly selective protein carriers within the membrane that bind to the transported substance, which could be an ion or a simple molecule, such as a sugar, an amino acid, or a nucleotide.
- These carrier proteins are called uniporters if they transport a single type of molecule and symporters or antiporters if they transport two different molecules together.
- Active transport is one of the most important functions of any cell.

### B. The sodium-potassium pump runs directly on ATP

- More than one-third of all of the energy expended by an animal cell that is not actively dividing is used in the active transport of sodium (Na+) and potassium (K+) ions.

### C. Coupled transport uses ATP indirectly

- Example: Transporting Glucose Glucose is such an important molecule that there are a variety of transporters for it, one of which was discussed earlier under passive transport.

**Key terms**

Active transport - one of the most important functions of any cell. It enables a cell to take up additional

Glucose - such an important molecule that there are a variety of transporters for it, one of which was discussed

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## 11. Transport Mechanisms (Active Transport and Bulk Vesicular Transport)

### BIOLOGY 1-HANDOUT11

### I. Bulk transport across the plasma membrane

### A. Exocytosis

- - The cell secretes certain molecules by the fusion of vesicles with the plasma membrane. - Many secretory cells use exocytosis to export products.

### B. Endocytosis

- - The cell takes in molecules and particulate matter by forming new vesicles from the plasma membrane.

### II. Types of Endocytosis

### 1. Phagocytosis

- - A cell engulfs a particle by extending pseudopodia (singular, pseudopodium) around it and packaging it within a membranous sac called a food vacuole.

### 2. Pinocytosis

- - A cell continually "gulps" droplets of extracellular fluid into tiny vesicles, formed by infoldings of the plasma membrane.

### 3. Receptor Mediated Endocytosis

- - A specialized type of pinocytosis that enables the cell to acquire bulk quantities of specific substances, even though those substances may not be very concentrated in the extracellular fluid.

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## 12. Structures and Functions of Biological Molecules (CHO, CHON, Enzymes and Nucleic Acids)

- Structures and Functions of Biological Molecules (CHO, CHON, Enzymes and Nucleic Acids)

### BIOLOGY 1-HANDOUT12

- Topic: Structures and Functions of Biological Molecules

### I. BIOLOGICAL MACROMOLECULES

- A polymer is a long molecule built by linking together a large number of small, similar chemical subunits called monomers.

### II. Carbohydrates

### They are loosely defined group of molecules that all

- Their empirical formula (which lists the number of atoms in the molecule with subscripts) is $(CH_2O)_n$ , where n is the number of carbon atoms.

### Example of Polysaccharide

- It is also for energy storage that enables us to do our daily activities. -- 1 of 6 -- Cellulose - a structural polysaccharide, also consists of glucose molecules linked in chains.
- Chitin - the structural material found in arthropods and many fungi, is a polymer of N-acetylglucosamine, a substituted version of glucose.

### III. Nucleic Acids

- -- 2 of 6 -- Nucleic acids carry information inside cells, just as disks contain the information in a computer or road maps display information needed by travelers.

### A. Deoxyribonucleic Acid

### B. Ribonucleic Acid

- As a carrier of information, the form of RNA called messenger RNA (mRNA) consists of transcribed single- stranded copies of portions of the DNA.

### IV. Proteins

### Polymers of Amino Acids

### A. Enzyme catalysis

### B. Defense

### C. Transport

### D. Support

- -- 4 of 6 -- Protein fibers play structural roles.
- These fibers include keratin in hair, fibrin in blood clots, and collagen.
- The last one, collagen, forms the matrix of skin, ligaments, tendons, and bones and is the most abundant protein in a vertebrate body.

### E. Motion

- Muscles contract through the sliding motion of two kinds of protein filaments: actin and myosin.

### F. Regulation

### G. Storage

### 1. Nonpolar amino acids, such as leucine, often have R groups that contain $-CH_2$ or $-CH_3$

### 2. Polar uncharged amino acids, such as threonine, have R groups that contain oxygen (or -OH)

- Some examples are methionine, which is often the first amino acid in a chain of amino acids; proline, which causes kinks in chains; and cysteine, which links chains together.

### V. Lipids

### Hydrophobic Molecules

- Fats consist of complex polymers of fatty acids attached to glycerol.

### Triglycerides- for energy storage

### Phospholipids- encloses the cell membrane

### Prostaglandin- chemical messengers for pain

### Steroid- for membrane and hormones

#### Key terms

Biological macromolecules - traditionally grouped into carbohydrates, nucleic acids, proteins, and lipids. In
Sugars - among the most important energy storage molecules, and they exist in several different forms
Polysaccharides - long polymers made up of monosaccharides that have been joined through
Enzymes - biological catalysts that facilitate specific chemical reactions. Because of this property, the
Calcium and iron - stored in the body by binding as ions to storage proteins
Lipids - insoluble in water because they have a high proportion of nonpolar C-H bonds
Fats - excellent energy-storage molecules. The energy stored in the C-H bonds of fats is more than

# 13. ATP and ADP Cycle

### ATP and ADP Cycle

### I. Introduction

### II. Metabolism

- Metabolism can be divided into two main categories: Catabolism includes all the reactions that break large molecules into smaller ones.
- Anabolism includes all the reactions that build larger molecules from smaller ones.

### III. Laws of Thermodynamics

### Energy is something you need to do work

### A. The first law of thermodynamics states that energy can't be created nor destroyed

- The first law of thermodynamics basically means that you can't make new energy, and you can't just make energy disappear.
- Any energy you use must come from somewhere and then go somewhere else when you are done.
- Moving energy from one place to another is called energy transfer.
- As energy is transferred from one place to another, it often changes from one type of energy to another.
- This process is called an energy transformation.
- For example, several forms of energy are involved in producing light from an electrical lamp: light energy, electrical energy, heat energy (from the warm bulb), plus whatever type of energy is used to produce the electrical energy in the first place, such as the flow of water in a hydroelectric plant.
- Because energy never disappears, but is just converted from one form to another, the first law of thermodynamics is also sometimes called the law of conservation of energy.

### WARNING

- In everyday language, conservation means to save.
- However, when scientists refer to conservation of energy, they do not mean saving energy - at least, not in the sense of driving a car that gets better gas mileage or something -- 2 of 5 -- along those lines.
- What scientists mean is that the total amount of energy in the universe always remains the same, or is saved.
- You can organize all the different types of energy into two main categories: Potential energy is energy that an object or molecule has because of its position or arrangement.
- Potential energy, also called stored energy, includes several types of energy: Chemical potential energy, such as that found in oil, natural gas, or food.
- One of the most important kinds of energy in cells is the chemical potential energy of food.

### WARNING

- You can easily confuse the scientific definition of potential, which means "stored," with the more common, everyday definition, which means "a possibility." In everyday language, you might say that an object has the potential to store energy, such as a spring with the potential to store energy when compressed.

### Kinetic energy

- The most important kinds of kinetic energy to cells are light, heat, and the movement of charged particles, such as ions and electrons.

### KEEP IN MIND

- The first law of thermodynamics says that energy can't be created or destroyed; it's just transferred from one form to another.
- The second law of thermodynamics states that chemical reactions that happen spontaneously increase the entropy, or disorder, of an isolated system.
- The second law of thermodynamics is complicated but powerful.

### The second law tells you that

- Heat never moves from a cooler object to a warmer object. -- 3 of 5 -- It's impossible to convert heat completely into work.

### III. Spontaneous and Non Spontaneous Reactions

### A. Spontaneous reactions

- In cellular biology, the most useful expressions of the second law of thermodynamics are those that explain which reactions in cells are spontaneous and which are not.

- The second law of thermodynamics is a very abstract idea, but it's an idea you have seen in action in your everyday life.

### B. Non Spontaneous Reactions

- The second law of thermodynamics says that the universe is constantly increasing in disorder.
- Entropy in the universe continues to increase, and the second law is satisfied.

**Key terms**

Cells - constantly converting matter and energy from one form to another in order to get what they need to stay
Metabolism - all of a cell's chemical reactions, such as breaking down food molecules or building up muscle
Work - moving something over a distance
Potential energy - energy that an object or molecule has because of its position or arrangement
Kinetic energy - energy that objects and molecules have because of their motion
Only after the spring - compressed and ready to do work does it have potential energy. Even though it's possible

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## 14. ATP and ADP Cycle

### ATP and ADP Cycle

### BIOLOGY 1-HANDOUT14

### I. Going to Work in Cellular Factory

- You know about the work you do in your life, but you may not be aware of all the work that is going on in your cells every day as you go about your business: Synthesis: Cells must build large, complex molecules, such as proteins and DNA.

### II. ATP: The Energy Currency of Cells

- The chief "currency" all cells use for their energy transactions is the nucleotide adenosine triphosphate (ATP).

### 1. Energy from food is transferred to ATP

### 2. ATP provides energy for cellular work

- Adenosine triphosphate (ATP) is a nucleotide, the same type of building block that is used to build DNA and RNA.
- It has three types of parts:

### The sugar ribose

### A nitrogenous base

### Three phosphate groups

### Energy coupling

- -- 1 of 4 -- Cells are constantly building and breaking down ATP, creating an ATP/ADP cycle.
- The breakdown of ATP to ADP and P requires a water molecule and is called ATP hydrolysis.
- A cell does three main kinds of work: 1.

### III. Enzymes: Biological Catalysts

- Although spontaneous reactions are energetically favorable and can occur without energy input from the cell, there is no guarantee on exactly when they will occur. -- 2 of 4 -- In order for a spontaneous reaction to happen, the reactants must find each other and then collide in just the right way and with enough kin...
- The amount of energy that is necessary to trigger the reaction is called the energy of activation (Ea).
- The breakdown of sugar is a spontaneous, exergonic reaction that releases usable energy to cells.

### Enzymes are important to cells because

- The functional groups on the amino acids that make up the enzyme interact with the reactants, altering chemical bonds so that the changes necessary for the chemical reaction are more likely to happen The combined effects of these interactions between enzymes and reactants (substrates) result in the lowering of the a...
- A good example of oxidation and reduction is the reaction between sodium (Na) and chlorine (Cl- ).
- Because oxidation and reduction reactions occur together - one molecule is oxidized, and one is reduced - these reactions are called redox reactions.
- Reduced means less, right, so how can gaining an electron make you reduced?
- The reason for this contradiction is because elements are compared to their most oxidized state.

### KEEP IN MIND

**Key terms**

Cells - constantly building and breaking down ATP, creating an ATP/ADP cycle. Cells store energy by building ATP
Life - very fast paced - too fast to just wait around for necessary chemical reactions to occur, even spontaneous
Sodium - oxidized and becomes the sodium ion (Na+). Chlorine is reduced and becomes the chloride ion (Cl)
Metabolism - all about getting what you need to stay alive. You know what you need on a big scale - good old

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## 15. Photosynthesis (Part 1)

### Photosynthesis (Part 1)

### I. Introduction

- Each day, the radiant energy that reaches Earth equals the power from about 1 million Hiroshima-sized atomic bombs.
- Photosynthesis captures about 1% of this huge supply of energy (an amount equal to 10,000 Hiroshima bombs) and uses it to provide the energy that drives all life

### II. Photosynthesis combines CO2 and H2O, producing glucose and O2

- These include a form of photosynthesis that does not produce oxygen (anoxygenic) and a form that does (oxygenic).
- These two types of photosynthesis share similarities in the types of pigments they use to trap light energy, but they differ in the arrangement and action of these pigments.

### II. Stages of Photosynthesis

### 1. capturing energy from sunlight

### 2. using the energy to make ATP and to reduce the compound NADH, an electron carrier, to NADPH

### 3. using the ATP and NADPH to power the synthesis of organic molecules from CO2 in the air

- The third stage, the formation of organic molecules from CO2, is called carbon fixation.

### III. Photosynthesis and Chloroplasts

- The internal membrane of chloroplasts, called the thylakoid membrane, is a continuous phospholipid bilayer organized into flattened sacs that are found stacked on one another in columns called grana (singular, granum).
- Surrounding the thylakoid membrane system is a semiliquid substance called stroma.

### IV. Light Dependent Reactions

- -- 2 of 4 -- Each pigment molecule within the photosystem is capable of capturing photons, which are packets of energy.

### IV. The Discovery of Photosynthetic Process

### O2 comes from water, not from CO2

- Third, they confirm that the ATP and NADPH from this early stage of photosynthesis are then used in the subsequent reactions to reduce carbon dioxide, forming simple sugars.

**Key terms**

Life - powered by sunshine. The energy used by most living cells comes ultimately from the Sun and is captured
Oxygenic photosynthesis - found in cyanobacteria, seven groups of algae, and essentially all land plants. These
Connections between grana - termed stroma lamellae. Surrounding the thylakoid membrane system is a

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## 16. Photosynthesis (Part 2)

### Photosynthesis (Part 2)

### I. Introduction

- The main reason is that they're autotrophs, meaning they can make their own food. (Autotroph literally means "self-feeding.") While you would slowly starve if not allowed to move from one spot, plants make all the food they need through photosynthesis.

### II. Plant Structures involved in Photosynthesis

- Phloem, tubes of cells that are specialized for sugar transport, move sugars from the green leaves (where the sugars are made) to the rest of the plant.

### III. Role of the Soil

- However, if you give them other means of support and fertilizer, plants will do just fine without soil.
- Minerals such as calcium, magnesium, and iron are necessary for the proper function of proteins.

- Some fertilizers are called plant food, but they're not really food!

## IV. The Sun

- The different types of radiation from the sun probably sound familiar to you - things like x-rays, ultraviolet radiation (the stuff that causes skin cancer), and microwaves.
- The different types of electromagnetic radiation travel from the Sun to the Earth in waves.
- Each type of electromagnetic energy is identified by its wavelength, the distance from one energy peak to another. (A nanometer is 1/100,000,000 of a meter, so wavelengths of light are pretty small!) The radiation captured by plants for photosynthesis has wavelengths between 400 and 700 nanometers.
- Note: All living things that do photosynthesis have some form of chlorophyll, as well as other photosynthetic pigments.

## Photosynthesis

## V. The Light Reaction

- Photosynthesis: $CO_2$ + $H_2O$ + light energy $C_6H_{12}O_6$ + $O_2$ It shows the transfer of matter (carbon dioxide and water) and energy (light) from the environment into the plant where they're combined to produce food in the form of carbohydrates (glucose), while oxygen is released back to the environment as waste.
- During photosynthesis, these little steps are organized into two separate sets of reactions in the plant: the light reactions and the Calvin cycle.
- In light reactions, plants capture light energy from the Sun and transform it into chemical energy in the form of the energy molecule ATP.
- To do this conversion, they need the molecules that were made during the light reactions: ATP provides chemical energy, and NADPH provides the necessary electrons.
- You can tell that NADPH is the reduced form because you can see that it's carrying a hydrogen atom at the end of its name. (Electrons are usually carried as part of hydrogen atoms.) The H in the name NADPH tells you that the electron carrier is bringing electron passengers to a new location.

## VI. Transferring Light Energy to Chemical Energy

- The cellular machine that makes this energy transfer possible is an electron transport chain. -- 3 of 6 -- The electron transport chain for photosynthesis essentially works the same way as the one that is used in cellular respiration.
- If you were the reaction center chlorophyll, you'd then pass one of these electrons to the electron transport chain.

## VII. Phosphorylation

- Photophosphorylation means making ATP using the energy from light.

## STEPS in NON CYCLIC PHOSPHORYLATION

## 2. The electrons from reaction center chlorophylls are excited and move to outer orbitals

- Chemiosmosis is used to make ATP.
- During this process, energy from the movement of electrons is used to transport hydrogen ions ($H^+$) across the thylakoid membrane.
- The hydrogen ions flow back across the membrane like water through a dam, and the energy of their movement is used to make ATP from ADP and inorganic phosphate.
- As light energy is absorbed by chlorophyll, some light energy is used to split water ($H_2O$) molecules by photolysis into protons ($H^+$), electrons ($e^-$), and oxygen gas ($O_2$).

## VII. The Calvin cycle

- NADPH is an important source of electrons for reactions that build molecules for the cell.
- The Calvin cycle transforms inorganic carbon in the form of carbon dioxide ($CO_2$) into organic carbon in the form of carbohydrates.

## STEPS IN CALVIN CYCLE

- Rubisco is a much-needed nickname for ribulose-1,5-bisphosphate carboxylase/oxygenase.
- This critical first step captures the carbon dioxide from the environment and is called carbon fixation.
- These 3-carbon molecules are called 3-phosphoglycerate.
- 3-phosphoglycerate is phosphorylated when it's given a phosphate group from ATP, which transfers energy from ATP to the 3-phosphoglycerate.
- Some G3P is used to make glucose and other sugars.
- This process is called biosynthesis, or just synthesis, because the plant is making sugars.
- Some G3P and ATP are used to regenerate RuBP, the 5-carbon sugar the plant started with.
- This step is called the regeneration phase.
- Therefore, the energy from the Sun, through the processes of photosynthesis and respiration, will be used to run your cells Kratz, R.F. .

**Key terms**

Plant structures - specialized for taking in the materials that they need:

Plant leaves - full of the green pigment chlorophyll, which is very good at absorbing blue and red light, but not

Soil - useful to plants because it gives them something sturdy to stand in, it may hold water for them, and it may

Photosynthesis - possible because the photosynthetic pigments that absorb the light are hooked up to a little

Photophosphorylation - really just another way of saying "the light reactions of photosynthesis."

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## 17. Cellular Respiration (Part 1)

### Cellular Respiration (Part 1)

### BIOLOGY 1-HANDOUT17

### I. Introduction

- The metabolic pathway that your cells use to rearrange food molecules in order to make them useful as building blocks or to capture their stored energy is called cellular respiration.

### II. What is Cellular Respiration?

- -It is the breakdown of food molecules such as carbohydrates, proteins, lipids and fats. -It is a metabolic pathway that, through a series of small steps, rearranges the atoms in the food molecules making the stored energy available to the cell.

### III. Types of Cellular Respiration

### A. Aerobic Respiration (with oxygen)

- -happens in mitochondria -reactants: glucose and oxygen -produces ATP, water and carbon dioxide -stages: glycolysis (anaerobic), Krebs cycle and oxidative phosphorylation. -can give large amounts of ATP

### B. Anaerobic Respiration (without oxygen)

- -happens in cytoplasm -reactant: glucose -products: ATP and lactic acid in animals; or ATP ethanol and CO2 (yeast) -stages: Fermentation -can give small amount of ATP

### IV. When does the energy transfer?

### Through many chemical reactions, your cells

- I include energy as a product as a reminder that this pathway is exergonic and thus makes energy available to the cell.

### V. Stages of Cellular Respiration

- Glycolysis is the first part of cellular respiration.
- The electron transport chain is a system of proteins in a membrane that use redox reactions to transfer energy to ATP.
- The final redox reaction is the transfer of electrons to oxygen, which is reduced to water.

**Key terms**

Glycolysis - first part of cellular respiration. During

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## 18. Cellular Respiration (Part 2)

### Cellular Respiration (Part 2)

### BIOLOGY 1-HANDOUT18

### I. Glycolysis

- The word glycolysis means "sugar splitting," and that is exactly what happens during this pathway.
- These smaller sugars are then oxidized and their remaining atoms rearranged to form two molecules of pyruvate. (Pyruvate is the ionized form of pyruvic acid.)

### Figure 1

### II. Citric Acid Cycle

### The citric acid cycle functions as a metabolic furnace

### Figure 2

### III. Chemiosmosis

- ATP is the most useful energy carrier for cells: Whenever a cell needs to do a job, it's usually ATP that provides the necessary energy.

### ATP?

- Electron transport chains are made of large protein complexes embedded in a membrane.

### Figure 3

- Most of the ATP that is made by cellular respiration is made by this protein in a process called the chemiosmotic theory of oxidative phosphorylation, or just oxidative phosphorylation for short.
- On one side of the membrane is the intermembrane space; on the other side is the matrix of the mitochondrion.

### IV. Fermentation

- The distinction between these two is that an electron transport chain is used in anaerobic respiration but not in fermentation. (The electron transport chain is also called the respiratory chain because of its role in both types of cellular respiration.) We have already mentioned anaerobic respiration, which takes p...
- Operation of the chain builds up a proton motive force used to produce ATP, but $H_2S$ (hydrogen sulfide). is made as a by-product rather than water.
- Fermentation is a way of harvesting chemical energy without using either oxygen or any electron transport chain-in other words, without cellular respiration.
- Remember, oxidation simply refers to the loss of electrons to an electron acceptor, so it does not need to involve oxygen.
- Overall, glycolysis is exergonic, and some of the energy made available is used to produce 2 ATP (net) by substrate-level phosphorylation.
- If oxygen is present, then additional ATP is made by oxidative phosphorylation when NADH passes electrons removed from glucose to the electron transport chain.

### Types of Fermentation

- Lactic acid fermentation- pyruvate is reduced directly by NADH to form lactate as an end product, regenerating $NAD^+$ with no release of $CO_2$. (Lactate is the ionized form of lactic acid.) Lactic acid fermentation by certain fungi and bacteria is used in the dairy industry to make cheese and yogurt.

#### Key terms

ATP - most useful energy carrier for cells: Whenever a cell needs to do a job, it's usually ATP that provides the
Electron transport chains - made of large protein complexes embedded in a membrane. They accept electrons
ATP synthase - embedded in the inner mitochondrial membrane right along with the electron transport chain. On
Fermentation - way of harvesting chemical energy without using either oxygen or any electron transport
Alcohol fermentation- pyruvate - converted to ethanol (ethyl alcohol) in two steps. The first step releases carbon
Lactic acid fermentation- pyruvate - reduced directly by NADH to form lactate as an end product, regenerating